UAV Hyperspectral Stereoscopic Remote Sensing Technique Supports NO_2 Localization With a Meter-Level Spatial Resolution

Haochen Peng, Chengzhi Xing, Tianyu Tang, Wei Tan, Yikai Li, Zhiguo Zhang, Qijin Zhang, and Cheng Liu

***Abstract*—Undetected meter-level pollution sources significantly increase health risks and remain difficult to identify and regulate. Satellite observations are constrained by coarse spatial resolution and cloud cover, while ground-based measurements rely on a single mode, offering only partial views of pollutant distribution. Both approaches fail to attribute fine-scale pollution to specific facilities. Here, we present a hyperspectral uncrewed aerial vehicle (UAV)-based 3-D remote sensing system that addresses these limitations, achieving temporal, spatial, and spectral resolutions of 15 s, 30 × 30 m, and 0.4 nm. Its lightweight 2-D spectral module expands observation modes from one (ground-based) to four, ensuring high accuracy in spectroscopy (R^2 = 0.996), concentration (R^2 = 0.927), and imaging (R = 0.868). Applied to NO_2 mapping in a kilometer-level urban area, the system traced emissions below 200 m altitude to an industrial site, with column densities exceeding 1.0×10^{16} molecules/cm^2. It further resolved meter-level NO_2 hotspots within satellite high-value pixels, identifying two facilities—an automotive assembly plant and an automotive testing plant—with column densities of 1.33×10^{16} and 1.26×10^{16} molecules/cm^2 and fluxes of 3.48 and 3.30 μg/m^2/s, respectively. Comparison with satellite imaging results underscores the necessity of fine-scale source attribution. The developed instrument demonstrates strong performance in accuracy, 3-D mapping, and meter-level source attribution and flux quantification, providing a new technical pathway for emission accountability and refined pollution control, thereby advancing atmospheric environmental management toward greater precision and effectiveness.**

***Index Terms*—Fine-scale source attribution, flux quantification, hyperspectral remote sensing, meter-level spatial resolution, stereoscopic observation, uncrewed aerial vehicle (UAV).**

I. Introduction

ATMOSPHERIC pollutants with complex sources and heterogeneous distributions pose a major threat to air quality and public health [1], [2], [3]. Nitrogen dioxide (NO_2) exemplifies this challenge. Long-term exposure to NO_2 is associated with increased health risks, with each 10 μg/m^3 increment raising the combined relative risk for multiple diseases above 1.05 [4], [5], [6]. NO_2 emissions originate from large stationary sources like coal-fired power plants [7], [8], as well as distributed sources such as vehicle exhaust [9], [10] and small industrial facilities [11], [12]. This mixed-source nature leads to unexpected high-concentration hotspots—near elevated roads in residential areas [13], [14], illegally discharging urban enterprises [15], unregulated suburban zones or [16]. Detecting such fine-scale features requires high-spatial-resolution observations and multiple measurements to reconstruct 3-D pollution patterns and accurately locate emission sources.

However, existing remote sensing approaches cannot simultaneously achieve high spatiotemporal resolution and 3-D coverage, hindering meter-level source attribution. Point sampling instruments offer rapid measurements but suffer from limited coverage and low efficiency, making fine-scale attribution challenging [17], [18]. Optical remote sensing also faces constraints: satellites are limited by orbital parameters, with the best temporal resolution at 1 h geostationary environment monitoring spectrometer (GEMS) [19], [20] and spatial resolution no finer than 2.1 × 4.4 km tropospheric emissions: monitoring of pollution (TEMPO) [21], [22] while also being affected by clouds [23], [24]. Furthermore, while recent algorithmic advancements (e.g., deep neural networks) can downscale satellite estimates to sub-100 m resolutions [25], these data-driven methods rely on statistical inference

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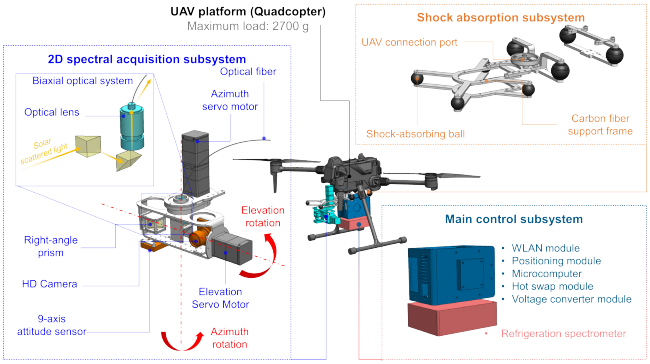


Fig. 1. 2-D hyperspectral imaging remote sensing UAV payload.

rather than direct physical observations, making their absolute accuracy difficult to guarantee. Additionally, even at the sub-100 m level, the spatial resolution remains insufficient for the precise identification of individual emission sources. Ground-based stations provide directional information but lack mobility to identify specific sources [26], [27], [28]. Vehicle-based platforms are route-constrained and unsuitable as standalone solutions [29], [30]. To overcome these limitations, new techniques are needed with high resolution, cloud/terrain resilience, and multimode observation. Uncrewed aerial vehicle (UAV)-based hyperspectral imaging presents such an opportunity [29], [31].

Airborne hyperspectral instruments typically require flight altitudes above 1 km to capture sufficient scattered light for trace gas retrieval, which inevitably degrades ground resolution [32], [33]. Moreover, bulky hyperspectral imagers that maintain both high spectral and spatial resolution are often unsuitable for UAV deployment [34], [35], [36]. While fixed-wing UAVs offer advantages in speed and range for fine-scale imaging, they require skilled operation and large takeoff/landing areas, restricting use in complex environments [37]. These constraints create inherent trade-offs among resolution, altitude, and payload. Xing et al. [38] recently made progress with a lightweight system achieving high resolution at low altitudes. However, their system remains limited to 1-D scanning and a single observation mode, preventing comprehensive 3-D pollution reconstruction.

This study presents a novel UAV-based hyperspectral system featuring a redesigned optical structure and 2-D payload capable of four complementary observation modes. We validate its spectral performance, imaging capabilities, and measurement accuracy across all modes within a unified 3-D space. The system enables high-resolution pollutant concentration mapping and emission flux quantification, achieving meter-level source attribution to bridge a critical gap in satellite monitoring and establish a new pathway for precise emission accountability.

II. Methods

A. Instrument Design

As shown in Fig. 1, the highly integrated, lightweight 2-D hyperspectral UAV payload (UAV payload) is mounted on a small quadcopter UAV platform with a maximum payload of 2.7 kg. The system consists of three main components: the 2-D spectral acquisition subsystem, the main control subsystem, and the shock absorption subsystem. Key performance parameters are listed in Table I.

The 2-D spectral acquisition subsystem captures 2-D solar scattered light signals. Its core components include servomotors, right-angle prisms, optical lenses, and optical fibers. Two servomotors drive two ultraviolet (UV)-coated quartz right-angle prisms, enabling independent rotation in azimuth and elevation directions. This allows 2-D scanning with a total solid angle coverage of up to π sr. The deflected light from the prisms enters a 0.5-in optical lens composed of two UV-enhanced coated lenses. Inside the lens, light converges onto the fiber end face and is transmitted through the optical fiber to the refrigeration spectrometer, converting the optical signal into an electrical signal. The azimuth scanning range is $-45°$ to $+45°$, while the elevation range is $-90°$ to $+90°$. The key feature of the 2-D spectral acquisition subsystem is that the optical fiber remains stationary while the prisms rotate. This design eliminates the mechanical resistance typically caused by fiber twisting. Therefore, smaller servomotors can be adopted, which directly reduces the payload's weight and size to ensure better flight stability. Additionally, the subsystem integrates a 3-D attitude sensor for real-time monitoring of the gimbal's absolute rotation angles and a high-definition camera module for live monitoring and recording of the observation area.

The main control subsystem offers higher integration and operational efficiency compared to fixed-site instruments. In addition to essential modules such as a microcomputer, wireless local area network (WLAN) module, positioning module, and voltage converter module, the conventional spectrometer with temperature control is replaced by the refrigeration spectrometer, effectively reducing size and weight. The UAV payload draws power directly from the UAV battery and incorporates a hot swap module, ensuring uninterrupted operation during battery replacement or UAV charging. Coupled with a UAV charging station, this enables automatic recharging of both platform and payload, thus advancing toward the realization of an unattended remote sensing station. The system also includes a wind velocity monitoring module to record real-time wind data during observation, supporting subsequent emission flux calculations. To bypass downwash interference, it retrieves ambient wind velocity by calculating the vector difference between GNSS-based ground speed and controller-derived air speed (detailed in Text S1 and Fig. S1).

The shock absorption subsystem consists of eight damping spheres and a carbon-fiber frame. It stabilizes the mounting of the 2-D spectral acquisition subsystem and main control subsystem on the UAV platform and effectively reduces high-frequency vibrations generated during flight.

B. Observation Mode

Thanks to the innovative design of the 2-D spectral acquisition subsystem, the UAV payload can perform nearly all observation modes available to current ground-based hyperspectral environmental sensing instruments, as well as some modes used by environmental monitoring satellites [20], [26], [28]. As shown in Fig. 2, four commonly used observation

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TABLE I
MAJOR PARAMETERS OF THE 2-D HYPERSPECTRAL IMAGING REMOTE SENSING UAV PAYLOAD

Modules	Parameters
Spectrometer	Spectral range: 300–420 nm Spectral resolution: 0.4 nm
Spectral collection lens	Field of view: 1.8° scattered light transmittance: >85%
HD Camera	Field of view: 45° Highest effective pixel: 3840(H)×2160(V)
Azimuth servo motor	Angles: -45°–45°, precision: 0.1°
Elevation servo motor	Angles: -90°–30°, 90°, precision: 0.1°
Operational performance	Flight endurance (with payload): 20 min Single mission duration: 6–18 min Operational altitude: 30–500 m
Overall performance	Total observed solid angle: π sr Weight: 2.3 kg Power consumption: 100 W Spatial resolution: 30 m×30 m Temporal resolution: 15 s

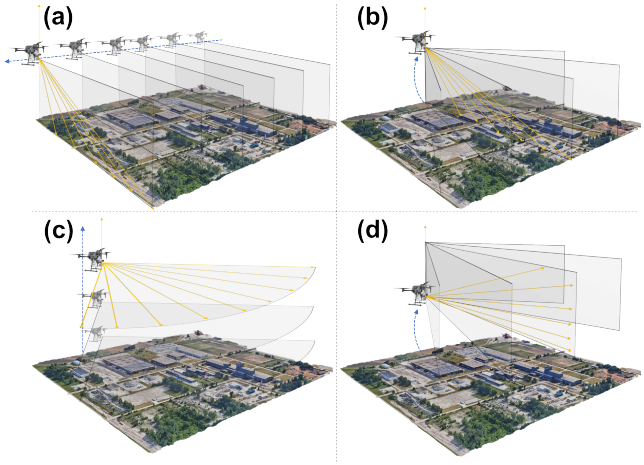


Fig. 2. Observation modes of UAV payload. (a) Ground grid scanning observation mode. (b) Ground sector scanning observation mode. (c) Horizontal stratified scanning observation mode. (d) Vertical profile scanning observation mode.

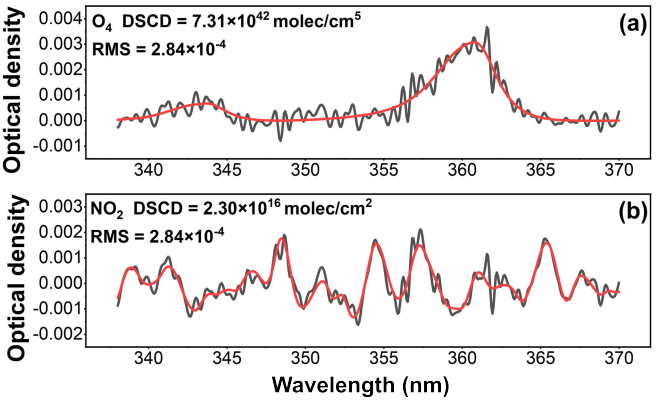


Fig. 3. Typical fitting examples for the ground scanning observation mode. The plots show the measured differential optical density (black lines), the fitted absorption features (red lines), and the corresponding root-mean-square (RMS) residual noise for (a) O_4 and (b) NO_2.

modes are demonstrated in this study: ground grid scanning observation mode, ground sector scanning observation mode, Horizontal stratified scanning observation mode, and vertical profile scanning observation mode.

As shown in Fig. 2(a), the ground grid scanning observation mode follows the observation strategy of previous payloads and consists of two steps: "UAV flight along predefined waypoints with hovering" and "elevation angle sequence scanning." After defining the observation area, the system automatically plans the hover points and the corresponding elevation sequences. The UAV sequentially flies to each preset location and performs elevation scans within a range of −5° to −50°. Upon completing a scan, the UAV moves to the next point until the entire observation area is covered. Each observation sequence also includes a 90° vertical reference spectrum acquisition. The flight altitude is set between 100 and 300 m. For the flight altitude of 100 m used in this study, the spatial resolution ranges from approximately 6–30 m (detailed resolution are listed in Table SI). This performance justifies the system's classification as achieving meter-level resolution. In this mode, the spectrometer's exposure time is typically 100 ms, with 150 exposures, and parameters can be dynamically adjusted based on weather conditions. Acquired spectra are processed following the method described in Text S2 to retrieve the differential slant column densities (dSCDs) of target trace gases. To illustrate the data quality and noise level, typical spectral fitting results for O_4, and NO_2 are presented in Fig. 3. According to Text S3, the air mass factor (AMF) can be calculated using either geometric or radiative transfer methods, allowing conversion of DSCD into vertical column density (VCD) and mapping of NO_2 concentration at each grid point in the observation area.

As shown in Fig. 2(b), the ground sector scanning observation mode adds azimuth rotation capability, enabling 2-D

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imaging while hovering. This mode minimizes errors caused by surface anisotropic reflectance that may occur in the ground grid scanning observation mode. For example, in the ground grid mode, the same surface may exhibit different reflectance at different hover points. Strong specular reflections from building glass can saturate the spectrometer. In contrast, the ground sector mode acquires data from a single hover point with 2-D rotation, effectively reducing such errors while also expanding coverage and reducing takeoff/landing time. Once the observation area is defined, the UAV payload sequentially measures according to the set azimuth sequence (typically −45° to 45°, step 5°) and elevation sequence (typically −5° to −50°). Exposure time, number of exposures, spectral analysis, and AMF calculations are consistent with the ground grid scanning observation mode.

As shown in Fig. 2(c), the horizontal stratified scanning observation mode is designed to capture the azimuthal distribution of pollutants at different altitude layers. UAV flight altitudes are set according to observation requirements and pollution levels, typically at 40, 80, and 120 m, with an azimuth sequence from −45° to 45° in 5° increments. Compared to ground scanning observation modes, this mode receives stronger optical flux. Under favorable weather conditions, exposure time is set to 50 ms with 300 exposures per point to enhance signal-to-noise ratio. Spectra are processed following Text S2 to retrieve the NO_2 DSCD, with typical fitting results shown in Fig. S2.

As shown in Fig. 2(d), the vertical profile scanning observation mode is dedicated to capturing the vertical distribution of atmospheric pollutants. Compared with fixed-site or vehicle-based multi axis differential optical absorption spectroscopy (MAX-DOAS), the UAV-mounted UAV payload can flexibly avoid ground obstacles such as trees, buildings, or walls, enabling unobstructed profile measurements at any altitude and location. The elevation sequence used is 1°, 2°, 3°, 4°, 5°, 6°, 8°, 10°, 15°, 30°, and 90°, while the azimuth sequence ranges from −45° to 45° with 15° increments. Acquired spectra are processed following Text S2 to retrieve trace gas DSCD (typical fitting results are shown in Fig. S3). Profile retrievals use the vector linearized discrete ordinate radiative transfer (VLIDORT) radiative transfer model (RTM) combined with the optimal estimation method (OEM) to infer vertical distributions of aerosols and trace gases. The retrieval is conducted in two steps: first, the vertical structure of aerosols is inverted using an exponential a priori profile; then, this structure is fixed as input for NO_2 and other trace gas vertical profile inversion. Analytic Jacobian matrices are used as weighting functions to improve algorithm efficiency, achieving approximately 100 m vertical resolution. The method accounts for smoothing errors, measurement noise, RTM approximation errors, absorption cross section uncertainty, and aerosol retrieval errors, with overall uncertainty assessed via Gaussian error propagation. This profile mode effectively captures the atmospheric vertical structure from near-surface up to the lower troposphere (0–4 km).

To guide practical application, we suggest selecting modes based on the specific scientific objective: ground grid scanning is best for high-precision mapping of large, unknown areas, while ground sector scanning offers rapid screening for specific facilities. Horizontal stratified scanning and vertical profile scanning are ideal for analyzing pollutant transport flux and vertical diffusion, respectively. For comprehensive monitoring, a strategic workflow is recommended: first, use ground modes to pinpoint hotspots, then deploy vertical and stratified modes to characterize the plume's height and transport pathway.

C. NO_2 Emission Estimation

As shown in (1), the flux divergence method was used in this study to estimate NO_2 emission fluxes at each grid point. This method calculates NO_2 flux by summing the NO_2 deposition term and diffusion term. Specifically, the deposition term is calculated from the NO_2 VCD (V) and NO_2 lifetime (τ), while the diffusion term is derived from the NO_2 VCD (V) and wind speed ($\vec{w}$) [39], [40]. The NO_2 lifetime (τ) was obtained from Xu et al. [8] calculation for the Hefei region during the corresponding period, and the wind speed ($\vec{w}$) was obtained from ERA5 (European Center for Medium-Range Weather Forecasts, https://cds.climate.copernicus.eu/cdsapp#!/dataset/) at 800–1000 hPa

$$E_{NO_2} = V/\tau + \nabla\left(V\vec{w}\right). \tag{1}$$

D. Auxiliary Data

The ground-based and vehicle-based MAX-DOAS instruments used in the validation experiments share identical optical system parameters, with a scanning accuracy of $< 0.1°$ and a field of view $< 0.3°$. The ground-based MAX-DOAS instrument was deployed in Hefei at the KXD site (39.91°N, 117.17°E), operating with an elevation sequence of 1°, 2°, 3°, 4°, 5°, 6°, 8°, 10°, 15°, 30°, and 90°, and a fixed azimuth angle of 46°, at a height of 15 m above ground level. The vehicle-based MAX-DOAS used an elevation sequence of 30°, 30°, 30°, 30°, 30°, and 90° to facilitate measurements of near-road atmospheric pollutant concentrations. High spatiotemporal resolution satellite data were obtained from the (GEMS, source: https://nesc.nier.go.kr), which provides 1-h temporal resolution and 7×8 km spatial resolution products. In this study, data from UTC 2:00 to UTC 6:00 were used, covering 32.14°N–32.18°N and 117.22°E–117.26°E. Tropospheric NO_2 raw data with cloud fractions exceeding 0.3 were filtered out prior to gridding [20].

III. Results and Discussion

A. Verification of Instrument Spectral Analysis

Optical remote sensing instruments typically verify their gas concentration measurement accuracy through concentration gradient experiments. As shown in Fig. S4, in this study, a laser driven light source (LDLS) combined with a collimation optical system produced quasi-parallel broadband white light. This stable, broadband source validates the instrument's intrinsic accuracy; due to the differential characteristics, the light source's specific spectral shape does not bias the retrieval. The beam passed through a 1 m long standard gas cell and

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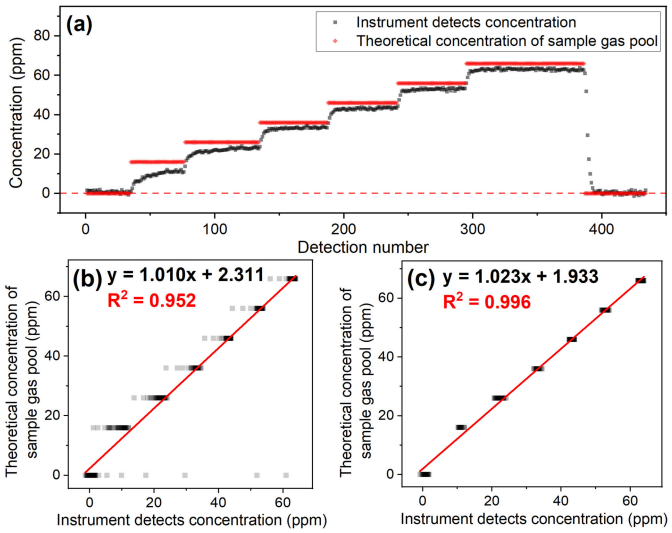


Fig. 4. Comparison (a) and (b) correlation between the concentration measured by the UAV payload and the concentration set by the standard gas pool. The (c) correlation between the concentration measured by the UAV payload under dynamic stability and the concentration set by the standard gas pool.

entered the optical system of the UAV payload. It is important to note that actual atmospheric observations typically involve optical paths of 5–10 km. To reproduce the same total optical absorption in the standard gas cell, the NO_2 concentrations were proportionally increased to the ppm level. This ensures the instrument detects signal strengths comparable to real-world urban monitoring. To prevent photolysis of NO_2 from affecting the results, the gas cell was covered with an opaque cloth during the experiment. The standard gas cell simulated an atmospheric NO_2 absorption layer, into which N_2/NO_2 mixed gases at specified concentrations (0, 16.6, 26.6, 36.6, 46.6, 56.6, and 66.6 ppm) were introduced via a gas mixing system. This transmission setup verified the instrument's spectral response accuracy under idealized conditions, eliminating external variables such as surface albedo. Each concentration was maintained for approximately 30 min. Spectral data acquired by the UAV payload were processed following the method described in Text S2 and Text S4 to retrieve absolute NO_2 concentrations. As shown in Fig. 4(a), the set concentrations (red circles) and measured concentrations (black squares) exhibit consistent stepwise trends, verifying the system's response accuracy. Because NO_2 in the gas cell requires time to reach dynamic equilibrium, measured values show transitions between gradients, with smaller concentration gradients requiring longer equilibrium times. The overall mean deviation was 2.46 ppm, primarily due to NO_2 adsorption losses and internal chemical reactions during gas flow. Fig. 4(b) shows linear fitting for all observation points ($R^2 = 0.952, y = 1.010x + 2.311$). As shown in Fig. 4(c), considering instability when switching gradients, further fitting was performed using data under dynamic equilibrium ($R^2 = 0.996, y = 1.023x + 1.933$), yielding improved results and demonstrating strong linear correlation between measured and set concentrations. This experiment confirms the UAV payload's high accuracy and sensitivity in spectral acquisition and analysis, enabling NO_2 remote sensing monitoring.

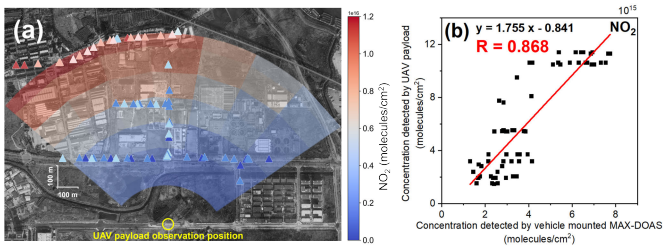


Fig. 5. Comparison (a) and (b) correlation between Ground sector scanning observation mode results and vehicle mounted observation results.

B. Verification of Instrument Profile Method

The UAV payload developed in this study provides both horizontal sector scanning and vertical profile observation functions similar to those of ground-based MAX-DOAS. Verifying consistency between the two in nonnegative elevation angle measurements of NO_2 concentrations is critical. In the experiment, the UAV payload and MAX-DOAS were placed side by side to ensure that their optical paths did not interfere. Both instruments were set with a 15 s integration time and identical elevation and azimuth sequences, observing from 10:00 to 17:00 on April 18, 2025. Due to differences in motor structure and control software, observation time and elevation angles were matched, selecting only the nearest time and identical elevation points for comparison. Linear fitting of the matched data yielded results of $y = 0.864x + 8.368, R^2 = 0.927$. As shown in Fig. S5, the NO_2 slant column densities (SCDs) measured by both systems are highly consistent, showing strong linear correlation. This indicates that the UAV payload achieves NO_2 measurement accuracy comparable to MAX-DOAS, providing a reliable basis for Horizontal stratified scanning observation mode and ground sector scanning observation mode.

C. Validation of Instrument Imaging Method

To verify the accuracy of the UAV payload's two ground scanning observation modes, this study referenced methods used to validate environmental satellite imaging. NO_2 VCDs observed by the UAV payload were compared with simultaneous vehicle-based MAX-DOAS measurements [37], [41]. The validation area and vehicle route are shown in Fig. S6. To minimize discrepancies caused by plume transport, both observations were performed synchronously from 13:00 to 14:00. As shown in Fig. 5(a), both instruments produced similar spatial NO_2 VCD distributions, with higher concentrations in the north and lower in the south. Due to dense vegetation along the road section, vehicle-based observations were partially blocked and discontinuous, while UAV observations were uninterrupted and continuous. Fig. 5(b) shows the linear fit between the two instruments ($y = 1.755x - 0.841$), with a correlation coefficient (R) of 0.868. The nonzero intercept is physically expected due to the different observation directions of the two instruments

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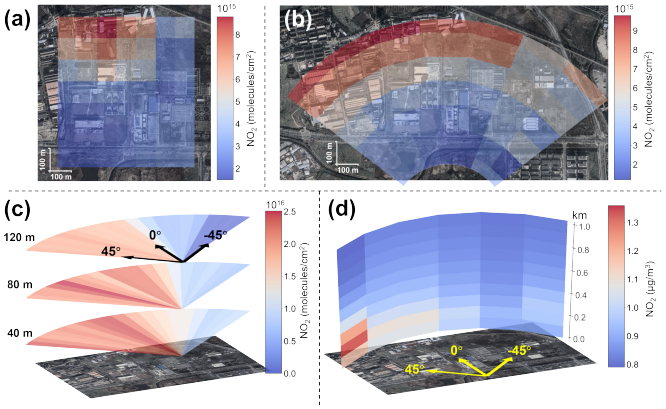


Fig. 6. NO_2 Imaging results of four observation modes. (a) Ground grid scanning observation mode. (b) Ground sector scanning observation mode. (c) Horizontal stratified scanning observation mode. (d) Vertical profile scanning observation mode.

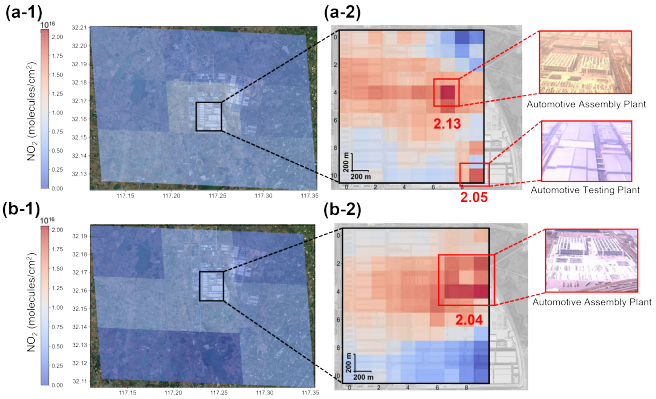


Fig. 7. Comparison of NO_2 imaging results between GEMS and the UAV payload. Panels (a-1) and (a-2) show the GEMS and UAV results on February 27, 2025, respectively. Panels (b-1) and (b-2) show the corresponding GEMS and UAV results on March 19, 2025.

(upward-viewing versus downward-viewing). These results demonstrate that the UAV payload achieves high imaging accuracy, providing a reliable foundation for ground grid scanning observation mode and ground sector scanning observation mode.

D. Observation of Trace Gas Concentration in Different Modes

Using the four observation modes, measurements were conducted in a selected region. Fig. S7 shows the satellite image of the study area, which includes industries such as transportation and engineering manufacturing industry, electronic information manufacturing industry, and biological manufacturing industry, as well as residential area, warehouses, and roads, forming a structurally complex area. Since the region 2 km north of the observation site consists mainly of fields or wasteland with no NO_2 emission sources, nearly all NO_2 plumes originated from within the observation area.

Fig. 6(a) presents the results from the ground grid scanning observation mode. Elevated NO_2 concentrations were observed around transportation and engineering manufacturing facilities located in the northern part of the study area, with a maximum value of 0.92×10^{16} molecules/cm^2. Compared with environmental monitoring satellites, the UAV payload operating at low altitude had a much shorter effective absorption path, covering only the near-surface atmosphere. As a result, the retrieved NO_2 concentrations were lower than the total column concentrations detected by satellites. In addition, the spatial resolution of this experiment reached the hectometer level. The grid size in Fig. 6(a) was 125×113 m, and it can be further refined depending on specific mission requirements.

As shown in Fig. 6(b), in the ground sector scanning observation mode, the UAV payload covered all industrial zones marked in Fig. S7. High NO_2 values were mainly associated with transportation and engineering manufacturing factories located in the north and northwest, where concentrations exceeded 1.0×10^{16} molecules/cm^2. Because the spatial resolution of this mode was slightly lower than that of the ground grid scanning observation mode, and the two observations were conducted at different times, some differences in concentration values occurred. The results demonstrate that this mode expands the coverage and improves observation efficiency while maintaining good data accuracy and spatial resolution.

Fig. 6(c) shows result from the horizontal stratified scanning observation mode. At the lowest layer (40 m), higher NO_2 concentrations were observed, primarily in the north (azimuth 5° and 10°) and northwest (azimuth 45°), with peak values above 1.32×10^{16} molecules/cm^2. At the top layer (120 m), concentrations were slightly lower but showed a diffusion pattern, with values ranging from 1.16×10^{16} to 1.22×10^{16} molecules/cm^2 in the northern and northwestern regions. This indicates that pollutant transport at this altitude was weaker than direct emissions from surface industries. These results demonstrate that this mode effectively captures horizontal variations of near-surface pollutants at different altitudes, making it suitable for identifying transport directions at specific layers and locating major emission sources.

Fig. 6(d) presents the results from the vertical profile scanning observation mode, showing NO_2 profiles between −45° and 45° within the 0–1 km altitude range. The results reveal a concentration peak below 200 m in the northwest direction, with a maximum value exceeding 2.30 μg/m^3. Leveraging the altitude flexibility of the UAV platform, this system overcomes the terrain dependence and limitations of traditional MAX-DOAS, enabling vertical profile observations at arbitrary locations and orientations. This provides an effective means for studying atmospheric vertical structures in complex terrain regions.

To demonstrate the system's multispecies detection capability, formaldehyde (HCHO) distributions were also retrieved. As shown in Figs. S8 and S9, the system achieved high-quality spectral fits and successfully captured HCHO patterns across all four modes, confirming its versatility in monitoring VOCs and other trace gases beyond NO_2.

E. Advantages of UAV Payload Over Satellite in High Spatial Resolution

The UAV payload, when operated in ground-based observation modes, demonstrates significant advantages over environmental monitoring satellites with low spatial resolution

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and ground-based instruments with limited viewing geometries. Fig. 7(a-1) and (b-1) shows GEMS observations of a newly constructed suburban industrial park on February 27 and March 19, 2025. Quantitative analysis (Table SII and Table SIII) confirms significant sub-pixel heterogeneity, with standard deviations of UAV-measured NO_2 reaching 1.75×10^{15} and 1.50×10^{15} molecules/cm^2 within the GEMS pixel on February 27 and March 19, respectively. This variability underscores the limitations of satellite resolution in capturing localized emissions. Due to its coarse spatial resolution, the industrial park is covered by only one to two satellite pixels, making it difficult to resolve individual emission sources within the park, let alone attribute emissions to specific factories or building units. In contrast, Fig. 7(a-2) and (b-2) presents UAV payload measurements in the ground grid scanning observation mode over the same area. On February 27, high NO_2 concentrations were mainly observed near the automotive assembly plant and the automotive testing plant, reaching 1.33×10^{16} molecules/cm^2 and 1.26×10^{16} molecules/cm^2, respectively. On March 19, the concentration hotspot was concentrated at the automotive assembly Plant, with a maximum of 0.80×10^{16} molecules/cm^2. Combined with on-site inspections and visual imagery from the onboard HD camera, it was found that both facilities experienced frequent heavy truck traffic during the observation periods, suggesting that transportation activities were likely a major source of NO_2 emissions. These results indicate that UAV-based remote sensing systems provide substantial advantages in identifying near-surface pollution sources, tracing emissions, and assigning responsibility, making them particularly suitable for kilometer-level emission monitoring and attribution studies.

Crucially, the drastic spatial gradients observed in Fig. 7 expose the substantial "spatial representativeness errors" [42] inherent in sparse ground networks, where stationary sites located even a few hundred meters from a source may fail to capture peak pollution levels. Therefore, we propose that the system's four observation modes, especially the ground-based modes which offer the highest spatial resolution, can serve as a critical pre-survey tool for optimizing the site selection of ground monitoring stations. By accurately mapping the horizontal distribution of pollutants and assessing the impact range of specific emission sources, the system can guide the establishment of new stations to ensure they are placed in representative locations, thereby minimizing monitoring blind spots.

Compared with traditional approaches that rely on satellite observations and ERA5 reanalysis wind data, UAV-based flux estimation offers clear advantages in both temporal responsiveness and spatial resolution. Fig. 8(a-1) and (b-1) show NO_2 emission flux distributions derived from GEMS data over the industrial park on February 27 and March 19, 2025. Under the higher wind speed of 6.4 m/s on March 19, the maximum emission flux reached 1.33 μg/m^2/s. Under the lower wind speed of 3.4 m/s on February 27, the maximum flux was 1.83 μg/m^2/s, higher than that of March 19. In both cases, the flux maxima were concentrated at the central grid cell, which was subsequently selected as the UAV observation region.

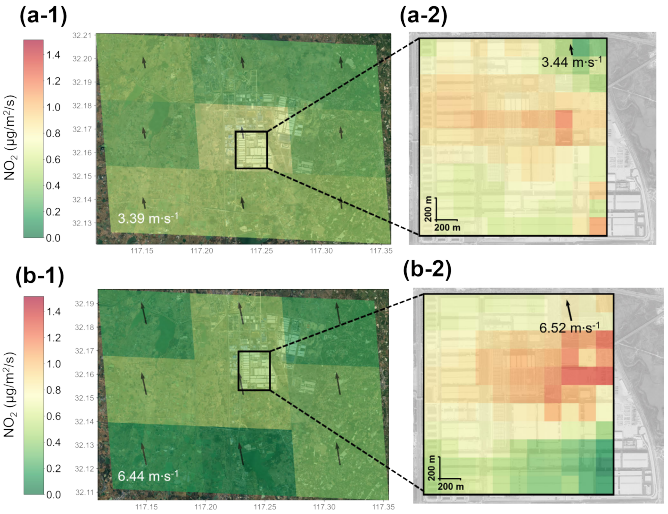


Fig. 8. Comparison of NO_2 emission imaging results between GEMS and the UAV payload. Panels (a-1) and (a-2) show the GEMS and UAV results on February 27, 2025, respectively. Panels (b-1) and (b-2) show the corresponding GEMS and UAV results on March 19, 2025.

Fig. 8(a-2) and (b-2) shows the emission flux spatial distributions retrieved by the UAV payload. On February 27, with an average wind speed of 3.44 m/s, the maximum fluxes in the automotive assembly plant and automotive testing plant areas were 3.48 and 3.30 μg/m^2/s, respectively. On March 19, with a higher average wind speed of 6.52 m/s, the automotive assembly plant exhibited fluxes exceeding 3.79 μg/m^2/s, surpassing the maximum flux observed on February 27. These findings highlight that UAV remote sensing with finer spatial resolution enables more accurate quantification of emission fluxes and allows the detection and characterization of emission sources at the meter level that cannot be resolved by satellites.

F. Control Errors of UAV and UAV Payloads

Benefiting from the design of the shock absorption subsystem, the UAV payload can effectively eliminate high-frequency vibrations induced by UAV flight during observations, as evidenced by the disappearance of the Jello effect in the camera integrated within the 2-D spectral acquisition subsystem. Through active closed-loop regulation of two servomotors installed on the UAV flight controller and the 2-D spectral acquisition subsystem, azimuthal, and pitch angle deviations caused by atmospheric turbulence are effectively suppressed, thereby maintaining stable pointing during measurements. However, due to payload weight limitations, the 2-D spectral acquisition subsystem is not equipped with active roll-angle compensation, and roll-angle adjustment relies entirely on the UAV's flight controller. To ensure both operational safety and observational accuracy, experiments must be conducted at wind speeds below 5.5 m/s (Beaufort scale 3), where instantaneous roll-angle deviations caused by gusts and turbulence remain smaller than $\pm 1°$/s.

The hovering accuracy of the UAV directly affects the positional accuracy of the imaging area. With a global navigation satellite system (GNSS) module, the hovering accuracy can reach ± 1.5 m horizontally and ± 0.1 m vertically. With

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a vision-based positioning module, the hovering accuracy improves to ±0.3 m in both horizontal and vertical directions. The UAV used in this study was equipped with both modules, achieving an actual hovering accuracy of ±1.5 m horizontally and ±0.3 m vertically. In future work, the integration of a real-time kinematic (RTK) module could further improve hovering accuracy from the meter level to the centimeter level [43], [44].

G. Algorithm Error of UAV Payload

Errors in the Vertical profile scanning observation mode can be divided into state vector retrieval errors and algorithmic errors. The sum of smoothing errors and DOAS fitting errors (noise errors) is generally considered the retrieval error of the state vector. The smoothing error arises from the limited vertical resolution of profile inversion, while the DOAS fitting error originates from uncertainties in spectral measurements. Algorithmic errors, on the other hand, arise from discrepancies between measured dSCDs and modeled dSCDs due to the inability of RTMs to perfectly reproduce the real atmospheric environment. Additional error sources, including uncertainties in absorption cross sections of trace gases, and retrieval errors introduced by aerosol inversion, have also been considered (see Text S5 and Table SIV).

IV. Conclusion and Prospects

In this study, a 2-D hyperspectral UAV payload was developed to achieve meter-level, 3-D monitoring of atmospheric pollutants. Comprehensive validation experiments—including spectral analysis, profile retrieval, and imaging verification—confirmed the accuracy of the system, demonstrating strong correlations with simulated environments (R^2 = 0.996), ground-based MAX-DOAS (R^2 = 0.927), and mobile measurements (R = 0.868). The system's four observation modes were successfully applied to a complex industrial area, where ground modes pinpointed NO_2 hotspots from manufacturing facilities with column densities exceeding 1.0×10^{16} molecules/cm^2, while stereoscopic scanning revealed a concentration peak (>2.30 μg/m^3) below 200 m in the northwest sector.

With a spatial resolution two orders of magnitude higher than satellites, the UAV payload effectively bridged the gap between regional monitoring and facility-scale attribution. It resolved emission sources to specific buildings (e.g., Automotive Assembly Plant) and identified heavy-duty traffic as a major contributor. Furthermore, by integrating onboard wind-field monitoring, the system refined flux quantification, measuring a maximum flux of 3.79 μg/m^2/s—significantly higher than satellite-grid estimates (1.33–1.83 μg/m^2/s). These results highlight the critical advantage of the UAV payload in combining fine-scale spatial observations with real-time wind data for precise pollution source attribution.

Looking forward, beyond integrating RTK modules to improve positioning accuracy from meter-level to centimeter-level, the primary focus will be on establishing a multi scale "Air-Space-Ground" integrated monitoring framework. In this framework, the UAV payload serves as the critical mesoscale link, bridging the gap between the broad coverage of satellites (e.g., GEMS, TROPOMI) and the high temporal resolution of ground-based networks. By combining satellite-identified hotspots with UAV-based fine-scale verification, this system offers a new pathway for environmental compliance monitoring. Beyond independent monitoring, this UAV system serves as a critical validation bridge for emerging sub-100 m satellite retrieval methodologies [25]. By providing meter-level spatial resolution data based on direct physical retrieval (DOAS), it effectively addresses the spatial representativeness errors inherent in point-based ground stations, creating a vital link in the "Air-Space-Ground" integrated environmental monitoring framework.

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